Functionalism

Putnam's functional states, multiple realisability, and the problem of qualia

This workbook consolidates what you have covered in the slideshow and handout. Work through it in order, or jump to the part you need. Every task can be done on your own, so if you missed a lesson you can still catch up. Answers to the recall tasks are at the back: try first, then check.

Name: _____ Date started: _____

In this workbook

1. Where am I now? A self-check
2. Key terms
3. Quick recall
4. Argument reconstruction
5. Objections and evaluation
6. Exam practice (3, 5, 12 and 25 marks)
7. Answers

> **Key Thesis**
> All mental states can be characterised in terms of their **functional roles**: their typical **inputs** (sensory stimuli), **outputs** (behaviour) and relations to **other mental states**. What a mental state *is* is what it *does*. Because a role can be played by different things, mental states can be **multiply realised**.

Part 01 Where am I now?

How to use this section

Tick a column for each statement **before** you start, then again **when you finish**. Green = I could explain it to someone else. Amber = I half get it. Red = not yet.

I can...	Start	End
define a function using input, output and the example of squaring	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
explain functionalism with the hardware/software analogy	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
explain why functionalism handles multiple realisability better than identity theory	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
explain why it does better than behaviourism (internal states, other mental states as outputs)	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
explain inverted qualia and Block's China thought experiment	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
apply the knowledge argument to functional facts	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
reach and defend a judgement on functionalism	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G

Part 02 Key terms

Task 1 · Match the term to its definition

Write the letter of the correct definition next to each term.

Term	Letter
Function	
Functional role	
Multiple realisability	
Realiser	
Inverted qualia	
China thought experiment	
Convergent evolution	

Term	Letter
Qualia	

	Definition
A	Unrelated species evolve similar features because they face similar functional demands.
B	The same functional role can be realised in different physical (or even non-physical) systems.
C	A process that takes a specific input (domain) and gives a specific output (range), e.g. 'the square of…'.
D	Two people functionally identical whose colour experiences are swapped, e.g. red for blue.
E	Whatever physically plays the functional role in a particular creature or system.
F	Block's case: the population of China duplicates a brain's functional organisation, apparently without one consciousness.
G	The causal job a mental state does: its typical causes, effects and links to other mental states.
H	Intrinsic, non-intentional phenomenal properties that are introspectively accessible.

Task 2 · Fill the gaps

Use the word bank. Each word is used once.

Key idea Word bank: inputs · Block · behaviour · does · hardware · organism · dualism · qualia

Functionalism says a mental state is defined by what it _____. It has sensory _____ and two kinds of output: _____ and other mental states. Putnam proposed that pain is a functional state of a whole _____. The _____/software analogy shows how different bodies can run the same mental 'program'. Functionalism is even compatible with _____. Critics such as Ned _____ argue a system could be functionally identical to us with no _____ at all.

Task 3 · In your own words

Using the jackal example, draw a simple input → state → output diagram for fear. Then explain why 'believing my life is in danger' counts as an output.

Part 03 Quick recall

How to use this section

Answer from memory first. Then check your notes, and add anything you missed in a different colour. That second colour shows you exactly what to revise.

1. What is a function? Give the maths example.

2. What two kinds of output can a functional state have?

3. What did Putnam hypothesise about pain?

4. Why is functionalism better than identity theory on multiple realisability?

5. Why is functionalism better than behaviourism?

6. How does convergent evolution support functionalism?

7. Explain the inverted qualia objection in one sentence.

8. What does Block's China thought experiment aim to show?

Part 04 Argument reconstruction

> **How to use this section**
> Each argument below has been jumbled. Number the lines into the right order (premises first, conclusion last). Then answer the questions underneath. Being able to set out an argument in standard form is exactly what the 5-mark and 12-mark questions reward.

Putnam's argument for functionalism

> *'I propose the hypothesis that pain, or the state of being in pain, is a functional state of a whole organism'*
> — Hilary Putnam, *The Nature of Mental States*

Order	Jumbled line
	So pain cannot be identical to any one type of physical state.
	Many different creatures, with very different physical make-ups, can be in pain.
	The best explanation of what they share is that pain is that functional role.
	What these creatures share is the role pain plays: caused by damage, causing avoidance and other mental states.
	Therefore, pain (and every mental state) is a functional state.

→ Which premise is the multiple realisability point against identity theory?

→ Why does premise 3 help functionalism avoid behaviourism's problems?

The inverted qualia objection

Order	Jumbled line
	If functionalism were true, functionally identical beings would have identical mental states.
	But it is possible that you experience what I would call red when you see the sky.
	Then we are functionally identical but differ in our mental states.
	Therefore, functionalism is false.

Order	Jumbled line
	You and I respond to the sky in exactly the same way: same inputs, same outputs, same links to other states.

→ How might a functionalist argue that premise 2 is not really possible?

→ Why do we never notice the inversion in real life, according to the thought experiment?

Part 05 Objections and evaluation

How to use this section

For each objection: explain the problem in your own words, write the best response, then judge whether the response works. The hint column is there if you are stuck; cover it up if you want a harder challenge.

Objection	Hint	The problem, in my own words	Best response	Does it work? Why?
Inverted qualia	Swapped red/blue; same words, same behaviour.			
Block's China thought experiment	A billion people with walkie-talkies duplicating a brain.			
The hard problem (Chalmers)	Why is any functioning accompanied by experience?			

Objection	Hint	The problem, in my own words	Best response	Does it work? Why?
Knowledge argument applied to function (spec)	Mary knows every functional fact about colour vision.			
Too liberal	If roles are all that matter, odd systems (a nation, a computer) could have minds.			
Compatible with dualism	Functionalism does not say what plays the role.			

Ranking the objections

Which objection is the **most** damaging, and which is the **least**? Justify your choice: a strong objection attacks a key premise and is hard to answer.

Most damaging: _____ because...

Least damaging: _____ because...

Building a judgement

Can functionalism explain all mental states? Consider whether it may succeed for beliefs and desires but fail for qualia.

Key idea Sentence starters: *The strongest argument for this view is... because...* · *However, this is undermined by...* · *A defender might reply that... but this fails / succeeds because...* · *Therefore, on balance...*

Part 06 Exam practice

How to use this section

On Paper 2 the **3-, 5- and 12-mark** questions test **AO1** only: knowledge and understanding, shown through precise, accurate explanation. The **25-mark** question is **5 marks AO1 + 20 marks AO2**: you must argue, evaluate and reach a judgement that your essay supports throughout.

Spend roughly a minute per mark. Plan before you write.

3 marks · What is meant by 'functionalism'?

Key idea AO1 · A precise definition or short point. One or two accurate sentences; no evaluation needed.

5 marks · Explain how functionalism accounts for the multiple realisability of mental states.

Key idea AO1 · Set out the idea or argument clearly and in logical order. Precise terminology; a standard-form layout often works best.

Planning frame

1. Define functionalism (role: inputs, outputs, other states).
2. Define multiple realisability.
3. Different realisers can play one role (hardware/software).
4. So the octopus and the human can both be in pain.

12 marks · Explain how Ned Block's China thought experiment challenges functionalism.

Key idea AO1 · A full, precise explanation. Show how the ideas connect logically, step by step. Do **not** evaluate: explain.

Planning frame

1. State functionalism precisely.
2. Describe the thought experiment: China duplicates the functional organisation of a brain.
3. Explain the intuition: there is no single consciousness, no qualia.
4. Show why this means function is not sufficient for mentality, so functionalism is false.

Extra notes

25 marks · Is functionalism the best account of the mind?

Key idea AO1 (5) + AO2 (20) · State your line of argument in the introduction and keep to it. Every paragraph should argue, not just describe: point → objection → response → mini-judgement. End with a conclusion that follows from what you have argued.

Model structure

Section	What to do	My plan
Introduction	Define the key terms. State your thesis: the answer you will defend.	
Paragraph 1	Multiple realisability: why functionalism beats identity theory → mini-judgement.	
Paragraph 2	Internal states: why it beats behaviourism → mini-judgement.	
Paragraph 3	Inverted qualia → functionalist reply → mini-judgement.	
Paragraph 4	Block's China / the knowledge argument → reply → mini-judgement.	
Conclusion	Restate your judgement and explain **why** the balance of argument supports it. No new arguments.	

Write your essay on lined paper, or continue below.

Reflect

Go back to Part 01 and fill in the **End** column. Which statement is still red or amber? Write one specific thing you will do about it.

Part 07 Answers

How to use this section

Use these to check Parts 02–04. Exam answers are not given here: bring them to your teacher, or mark against the AO guidance.

Task 1 · Matching

Function = C · Functional role = G · Multiple realisability = B · Realiser = E · Inverted qualia = D · China thought experiment = F · Convergent evolution = A · Qualia = H

Task 2 · Gaps

(1) does · (2) inputs · (3) behaviour · (4) organism · (5) hardware · (6) dualism · (7) Block · (8) qualia

Quick recall

1. A process taking a specific input to a specific output; e.g. squaring: input 2 → output 4.
2. Behaviour (e.g. running) and other mental states (e.g. believing my life is in danger).
3. That pain is a functional state of a whole organism.
4. Pain is defined by role, not brain chemistry, so animals or aliens with different brains can share it.
5. It allows internal states and links between mental states, so pain need not always produce pain-behaviour (e.g. staying quiet to avoid a mob).
6. Different species realise the same function with different physical structures (e.g. octopus and human eyes).
7. Two people could be functionally identical yet have swapped colour experiences, so mental states are not just functional roles.
8. Perfect functional duplication without consciousness is possible in principle, so function is not sufficient for mind.

Arguments in order

Putnam's argument for functionalism

1. Many different creatures, with very different physical make-ups, can be in pain.
2. So pain cannot be identical to any one type of physical state.

3. What these creatures share is the role pain plays: caused by damage, causing avoidance and other mental states.
4. The best explanation of what they share is that pain is that functional role.
5. Therefore, pain (and every mental state) is a functional state.

The inverted qualia objection

1. You and I respond to the sky in exactly the same way: same inputs, same outputs, same links to other states.
2. But it is possible that you experience what I would call red when you see the sky.
3. Then we are functionally identical but differ in our mental states.
4. If functionalism were true, functionally identical beings would have identical mental states.
5. Therefore, functionalism is false.

Objections: model responses

Objection	Possible response
Inverted qualia	Some functionalists deny genuine inversion is possible; others accept qualia are not functional but say beliefs and desires still are.
Block's China thought experiment	Bite the bullet: the system might be conscious; our intuition reflects scale and speed, not a real absence.
The hard problem (Chalmers)	Functionalists may say this is a pseudo-problem, or that the hard problem is a problem for every theory.
Knowledge argument applied to function (spec)	Use the replies to Jackson: ability, acquaintance, or new knowledge/old fact.
Too liberal	Specify roles more finely (e.g. at the level of a whole organism, as Putnam does).
Compatible with dualism	This can be a strength (neutrality), but means it does not by itself establish physicalism.